

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.48008	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.6016	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1478.55462	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.273435	0.273435	138.4686	0.0	0.0

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 43.48008.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.

- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.6016.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1478.55462.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 138.4686.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Solver-Evolution Suite Summary

Condition	Feasibility Rate	Heldout Feasible Gap	Heldout Runtime (ms)	Robustness (Dispersion)
Baseline Nearest Neighbor	1.0	0.2734	43.48	0.0028
Baseline Clarke-Wright Savings	1.0	**0.0738 (best)**	77.60	0.0078
Baseline Regret Insertion + LS	1.0	0.4664	1478.55	0.0869
Solver Evolution (phase9)	1.0	0.2734	138.47	0.0028

- ****Feasibility:**** All conditions, including the solver evolution, achieved perfect feasibility (1.0).
- ****Objective Gap:**** The solver evolution matched the nearest neighbor baseline (0.2734) but did not improve on the best baseline, Clarke-Wright Savings (0.0738).
- ****Runtime:**** Solver evolution's runtime (138.5 ms) is higher than nearest neighbor (43.5 ms) and Clarke-Wright (77.6 ms), but dramatically faster than regret insertion local search (~1479 ms).
- ****Robustness:**** Solver evolution matches nearest neighbor's low robustness dispersion (0.0028), indicating stable performance across held-out structure families.
- ****Training vs. Held-out:**** No significant generalization or improvement since training and held-out gaps match closely in all cases.

Conclusion

The evolved solver did ****not**** clearly beat the fixed baselines. It replicates the nearest neighbor baseline's performance in feasibility, gap, and robustness but falls short of Clarke-Wright Savings in objective gap, with a higher runtime.

****Summary:**** The phase-9 solver evolution maintains feasibility and stability but does not surpass classical baselines in solution quality or efficiency on held-out test families.